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SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

I G. Hendo (192521215) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data centers, and distributed servers.
2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

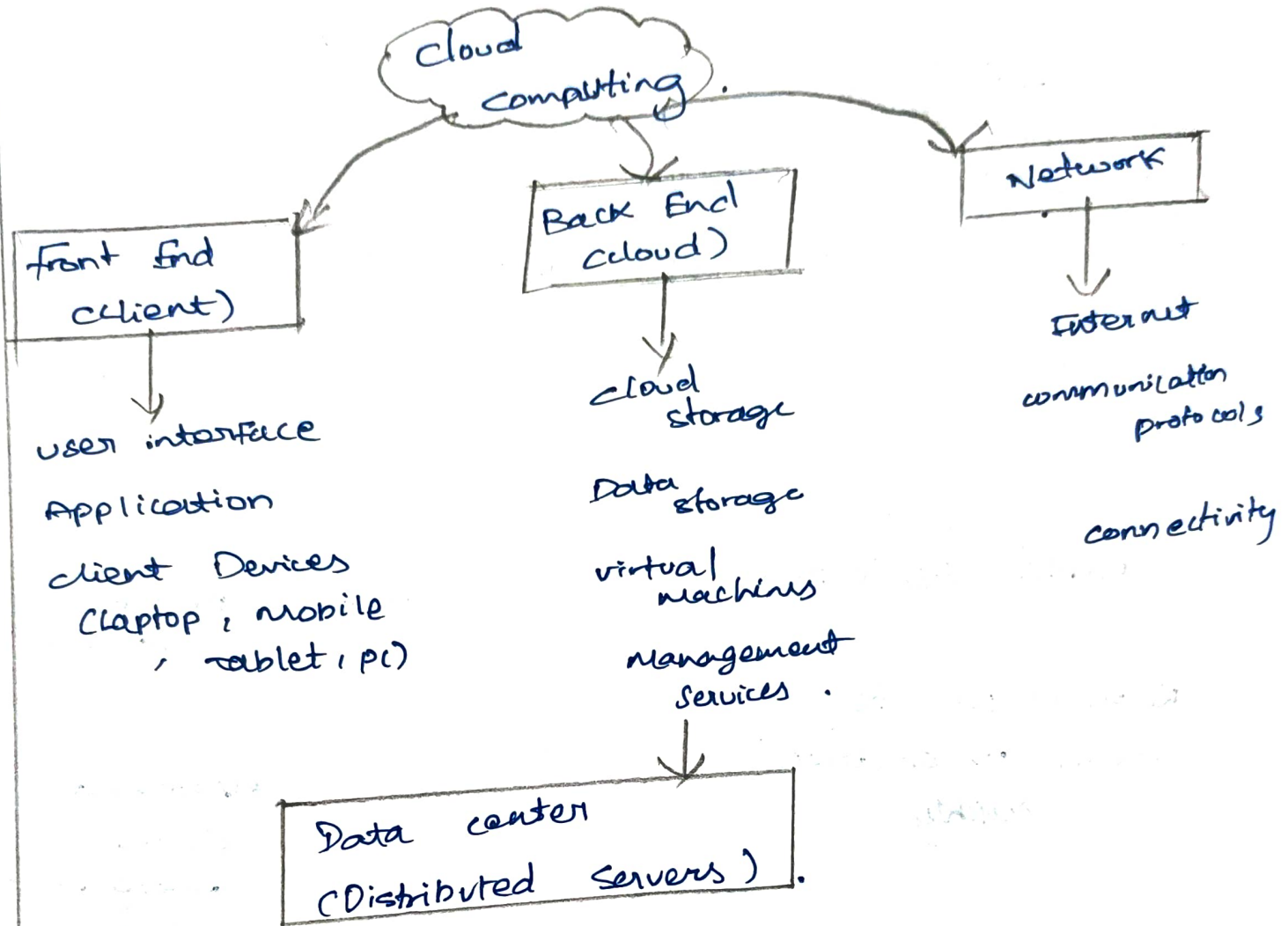
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CSA-1521

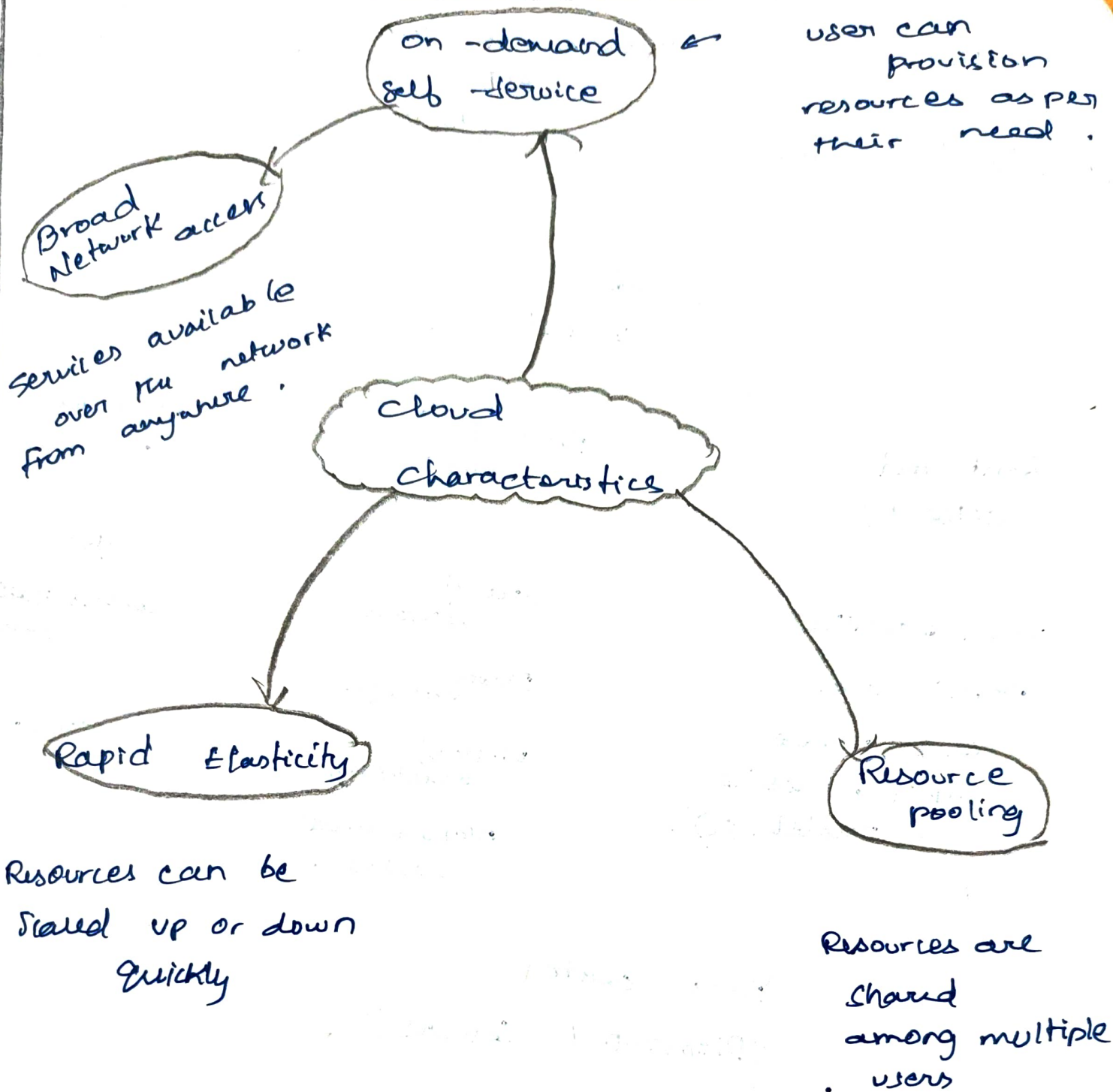
Assessment Tool AT-2

Gi. Harandra
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Cloud
computing

1) cloud computing Architecture.



2) Characteristics of cloud computing?



Deployment models:-

cloud Deployment models

public cloud

services provided
over the internet
for public use.

Ex:- AWS, Azure
google cloud.

private cloud

used by a
single organization
Greater control
and security

hybrid cloud

Combination of public
and private clouds.
Provides flexibility.

community
cloud.

used by a specific
community with
common concerns.

Ex:- Govt.
organizations.

4) Cloud Service models :-

cloud service models

IaaS
(Infrastructure
as a Service)

provides virtualized
computing resources
over the internet.
Ex:- AWS, EC2.

PaaS
(Platform as a
Service)

provides platform
and environment
to develop, test and
deploy applications.
Ex:- Google App Engine.

SaaS
(Software as a
Service)

provides
software apps
over the
internet.
Ex:- Gmail,
Microsoft
365.

5) Benefits and challenges :-

Benefits

- Cost Saving
- Scalability
- Flexibility
- High Availability
- Disaster Recovery
- Automatic updates
- Global Access.

Challenges

- Security concerns
- Downtime
- Data Privacy
- Vendor Lock-in
- Compliance Issues
- Internet Dependency.